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Design a differential amplifier operating on a lower supply voltage of 1.8 V, utilizing a wide-swing (high-compliance) cascode current mirror load. Analyze the trade-offs made to maximize the output voltage swing while maintaining a minimum voltage gain of 40 dB

ANALOG IC Design-Mini Project

MANISH

OBJECTIVES

Primary Objectives

- Design a differential amplifier operating on a lower supply voltage of 1.8 V.
- With a wide-swing cascode current mirror as load.
- Maintaining a minimum voltage gain of 40 dB.
- Analyzing the trade-of made to maximize the output voltage swing.
- Maintaining all Transistor in saturation mode.

Technical Motivation

- Standard cascode mirrors consume high voltage headroom (Vth+2Vov), limiting overall dynamic range.
- Wide-swing cascode biasing lowers required minimum voltage headroom to just 2Vov.

FEATURES

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- High DC Voltage Gain

Achieves high single-stage gain by utilizing cascode output impedance boosting without needing extra amplifier stages.

- Wide Output Swing

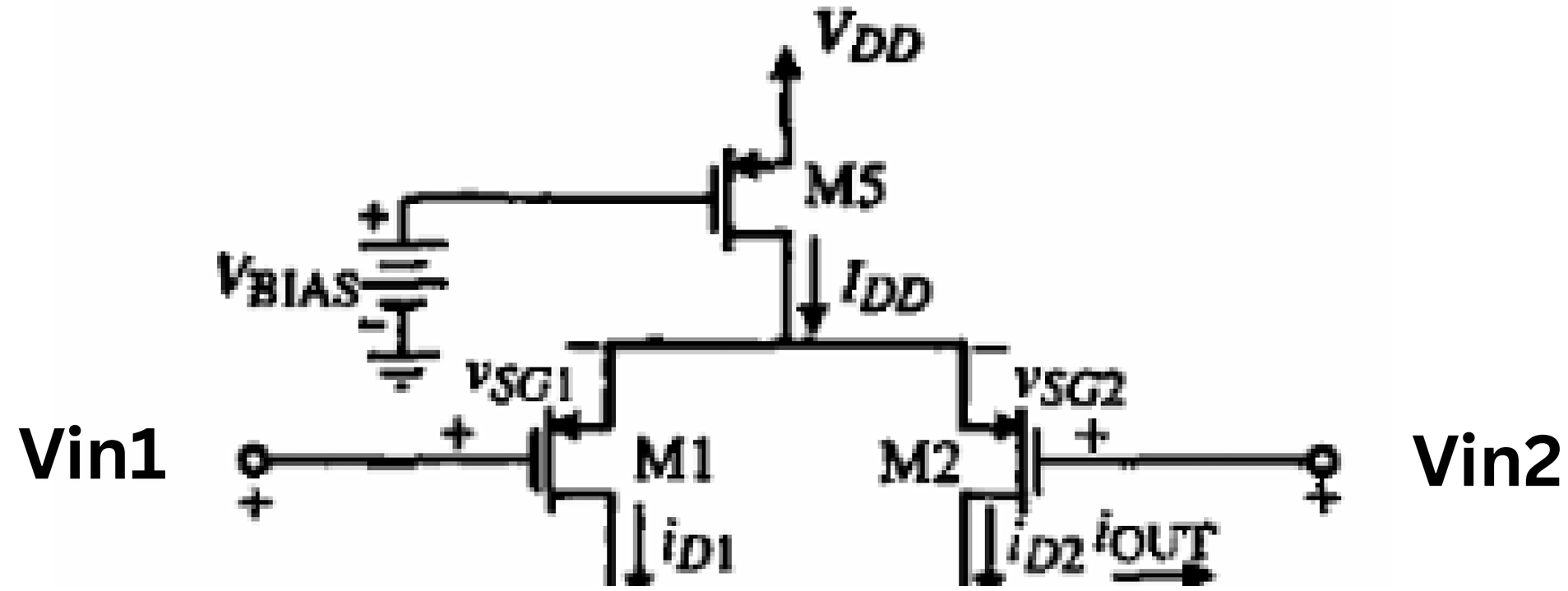
Wide-swing active load enables maximum output dynamic range, bounded by minimum compliance headroom of above ground.

- Noise & CMRR

PMOS differential input pair provides low noise and high Common-Mode Rejection Ratio (CMRR) against supply fluctuations.

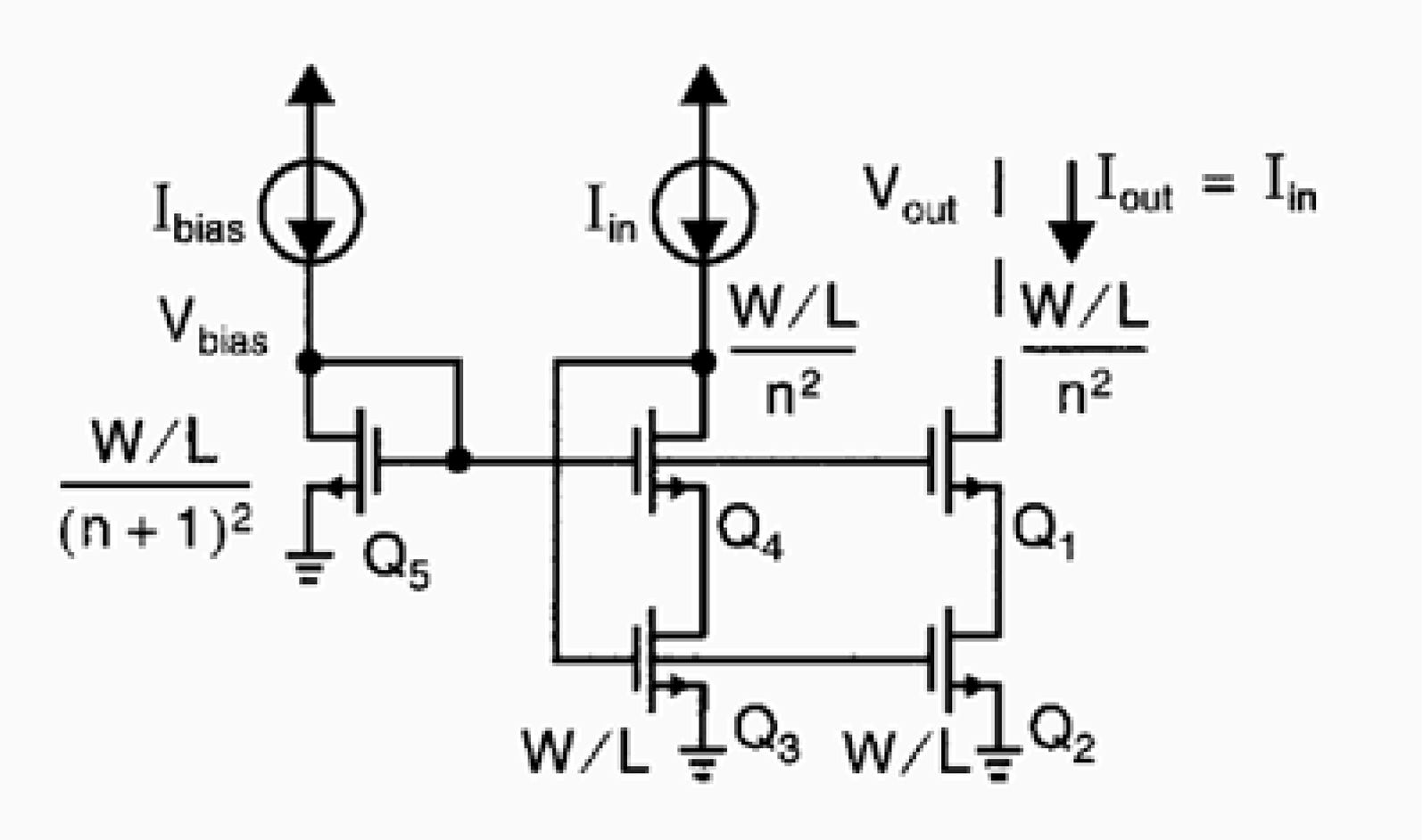
SCHEMATIC

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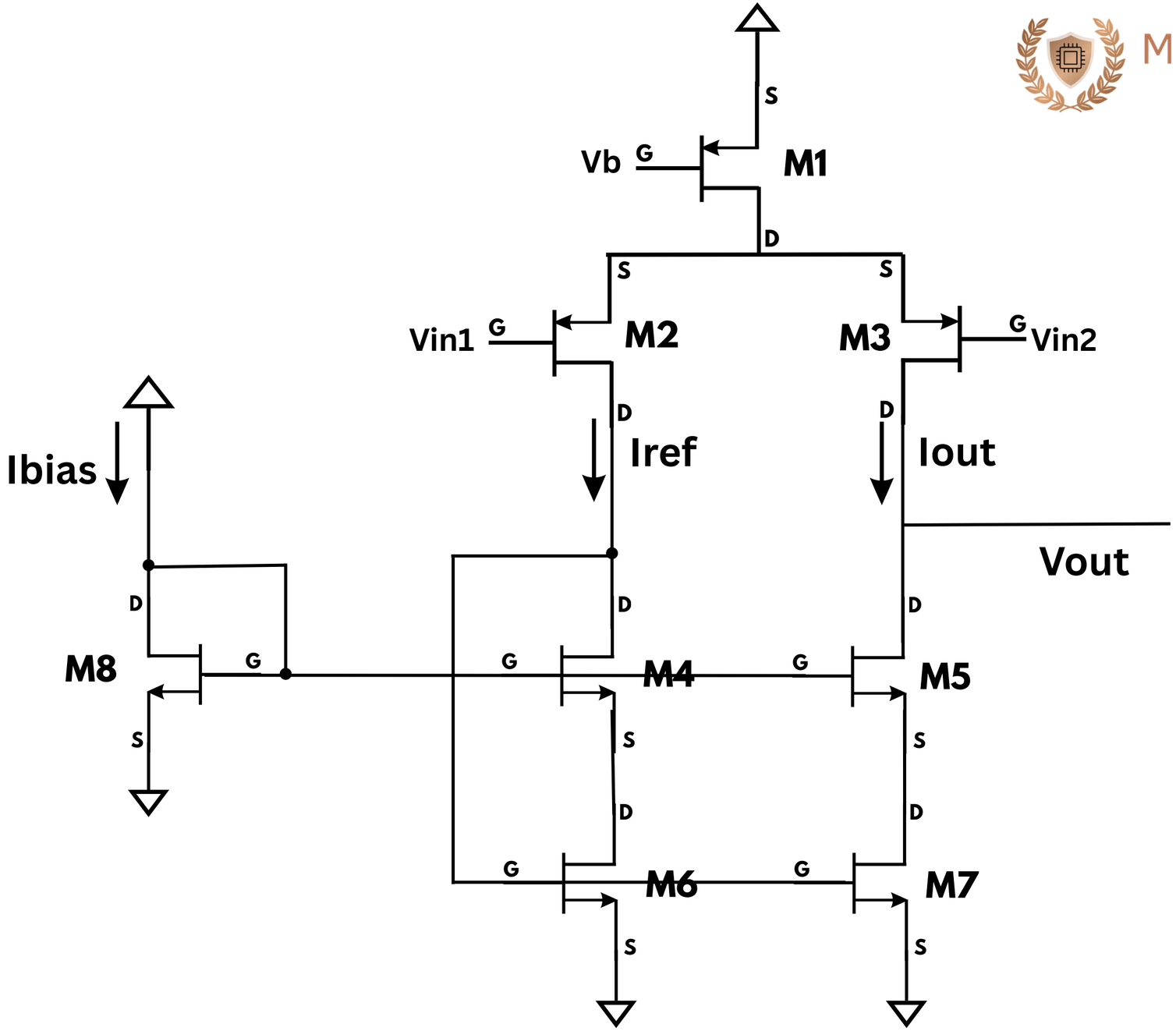


SCHEMATIC

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DESIGN


S
Vb
G
M1
D
S
S
Vin1
G
M2
M3
G
Vin2
D
D
Ibias
Iref
Iout
Vout
D
D
D
M8
G
G
G
M4
M5
S
S
S
D
D
G
G
M6
M7
S
S

WORKING

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- **Differential Current Steering**:

When the input voltage on one branch rises relative to the other, the circuit doesn't create new current it simply balance the current into one branch and away from the other. So like if left PMOS gets more current (I_tail / 2 + Δ i), the right PMOS automatically gets less (I_tail / 2 - Δi) .

- **Active Load Current Mirroring**:

The right output node has two opposing forces:

The top PMOS branch trying to push its current (i_d2) into the output.

The bottom NMOS load pulling the mirrored current (i_d1) down to ground.

Because the two currents move in opposite directions (+Δ i vs -Δi), they push and pull against the node's high output resistance, converting current difference into a huge single-ended voltage swing

- **Cascode Impedance Shielding**:

A cascode transistor is stacked directly on top of the bottom mirror transistor. It acts as an impedance multiplier and voltage buffer. It isolates the bottom transistor's drain, holding that node virtually constant regardless of how wildly Vout fluctuates.Because the bottom transistor sees a stable voltage, its current is constant.

- **Headroom Preservation**:

Wide-swing gate bias holds the bottom mirror drain-source voltage at exactly , preventing premature saturation loss. and makes Vds= Vov.

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DESIGN

Assumed Vov=0.15V, MnCox = 165µ A/V², Vthn = 0.53V, Vthp = 0.43V, MpCox = 45µ A/V²

Iss=20µA and VDD = 1.8V, Av=40db

From the top Transistor M1 , the (W/L)1 is ,

$$\frac{2\,(Id)}{MpCox \times Vov^2} = \frac{2\left(20 \times 10^{-6}\right)}{45 \times 10^{-6} \times (0.15)^2} = \boxed{39.5 \approx 40}$$

Vb is ,

$$Vsg = Vov + Vthp \qquad Vs - Vg = Vsg$$

$$Vg = Vb = Vs - (Vov + Vthp) \qquad Vb = 1.8 - 0.15 - 0.43 \qquad \boxed{Vb = 1.22V}$$

For the differential Pair Transistor M2 & M3 , the (W/L)2-3 is ,

$$\frac{2\,(Id)}{MpCox \times Vov^2} = \frac{2\left(10 \times 10^{-6}\right)}{45 \times 10^{-6} \times (0.15)^2} = \boxed{19.75 \approx 20}$$

For the main current mirror M4 & M5 , and the cascade mirror M6 & M7 the (W/L) is ,

n = 1

$$\frac{2\,(Id)}{MnCox \times Vov^2} = \frac{2\left(10 \times 10^{-6}\right)}{165 \times 10^{-6} \times (0.15)^2} = \boxed{5.38 \approx 5}$$

For the Diode connected Transistor M8 , the (W/L)8 is ,

$$\frac{(\frac{W}{L})}{(n+1)^2} = \frac{(5)}{(1+1)^2} = \boxed{1.25}$$

For the Output Voltage ,

$$Vout, min = 2Vov = 2 \times 0.15 \quad Vout, min = \boxed{0.3V}$$

$$Vout, max = VDD - 2Vov = 1.8 - 0.3 \quad Vout, max = \boxed{1.5V}$$

$$Vout, swing = Vout, max - Vout, min$$

$$Vout, swing = \boxed{1.2V}$$

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For the Input Voltage Vin,

$$Vin, max = 2Vov + Vthp \qquad = 2 \times 0.15 + 0.43 \qquad Vin, max = \boxed{0.73V}$$

$$Vin, min = VDD - 2Vov - Vthp \qquad = 1.8 - 0.3 - 0.43$$

$$Vin, min = \boxed{1.07V}$$

For the Voltage gain ,

$$Av = \frac{Vout}{Vin, diff} = -Gm.\, Rout$$

$Gm = gm, in = gm, diff$ (Transconductance of input diff. pair transistor)

$$Rout = Rup \| Rdown$$

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$$Rup = ro, in = ro, diff$$

(In small - signal analysis gate and source terminal of M1 are AC ground
so vsg=0, only M2 & M3 will be considered for pull-up impedence)

$$Rdown = gm5ro5.ro7$$

$$Rdown = gm.ro^2$$

(For pull-down impedence its the cascade current mirror design)

$$Av = -Gm.Rout$$

$$Av = 40db \quad = 10^{\frac{40}{20}} = 100$$

$$Gm = gm, in = gm, diff = \frac{2Id}{Vov} = \frac{2\left(10 \times 10^{-6}\right)}{0.15} = \boxed{133.34 \times 10^{-6} S}$$

$$Rout = \frac{Av}{Gm} = \frac{100}{133.34 \times 10^{-6}} = 749.96K\Omega \approx \boxed{750K\Omega}$$

$$Rdown \gg Rup$$

$$Rout \approx Rup$$ (Because *Rup* dominates *Rout*)

$$Rout = Rup = ro, diff = r2 = r3 = \boxed{750K\Omega} \qquad r1 = \frac{1}{0.1333 \times 20 \times 10^{-6}} = \boxed{375K\Omega}$$

$$\lambda = \frac{1}{r0.Id} = \frac{1}{750K \times 10 \times 10^{-6}} = \boxed{0.1333}$$

$$r4 = r5 = r6 = r7 = \frac{1}{0.1333 \times 10 \times 10^{-6}} = 750K\Omega \qquad r8 = \frac{1}{0.1333 \times 10 \times 10^{-6}} = 750K\Omega$$

$$gm, load = gm4 = gm5 = gm6 = gm7 = gm8 = \frac{2\,(Id)}{Vov} = \frac{2 \times 10 \times 10^{-6}}{0.15} = 133.34 \times 10^{-6} S$$

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$$Rdown = gm,load \times r0^2 = 133.34 \times 10^{-6} \times (750K)^2 = \boxed{75M\Omega}$$

$$Av,cm = \frac{Vout}{Vin,cm} = -\frac{1}{2 \times gm,load \times Rtail} \qquad Rtail = rtail = r1 = 375K\Omega$$

$$Av,cm = \frac{1}{-2 \times 133.34 \times 10^{-6} \times 375K} = -9.999 \times 10^{-3} = \boxed{-40db}$$

$$CMRR = |\frac{Av,diff}{Av,cm}| = 40db + 40db = \boxed{80db}$$

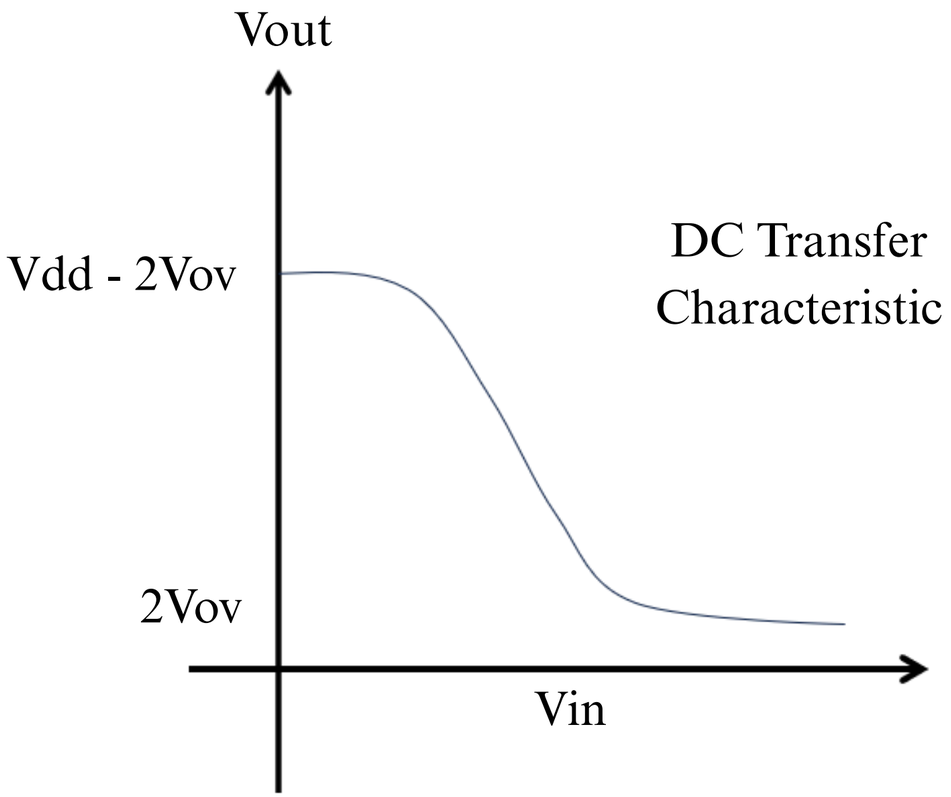
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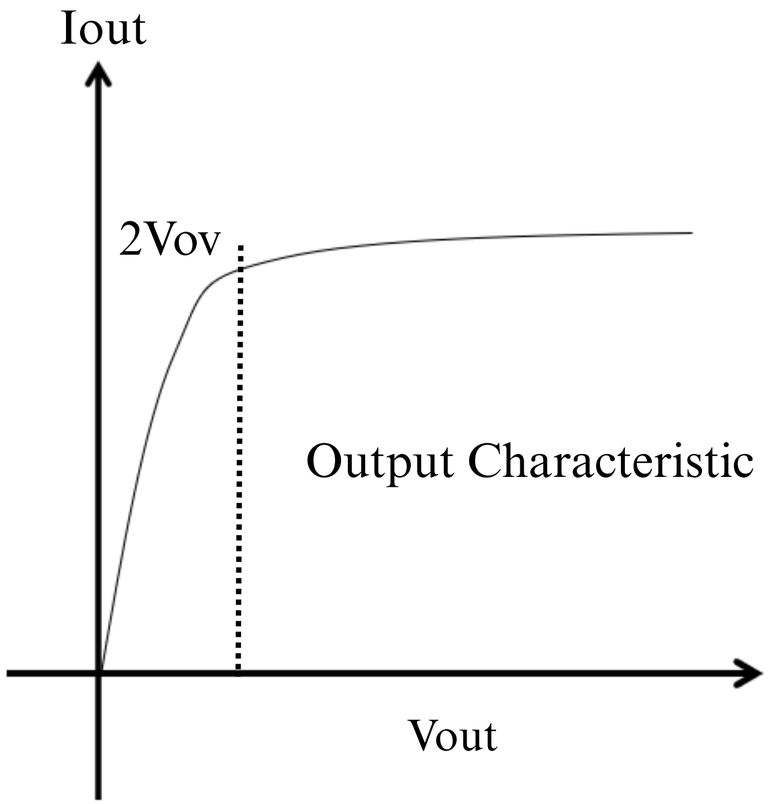
DESIGN SPECIFICATION

Transistor	Type	Id	Vov	W/L	W	L	ro
M1	PMOS	20µ	0.15V	40	8 X 50	10	375K
M2	PMOS	10µ	0.15V	20	8 X 25	10	750K
M3	PMOS	10µ	0.15V	20	8 X 25	10	750K
M4	NMOS	10µ	0.15V	5	25	5	750K
M5	NMOS	10µ	0.15V	5	25	5	750K
M6	NMOS	10µ	0.15V	5	25	5	750K
M7	NMOS	10µ	0.15V	5	25	5	750K
M8	NMOS	10µ	0.15V	1.25	6.25	5	750K

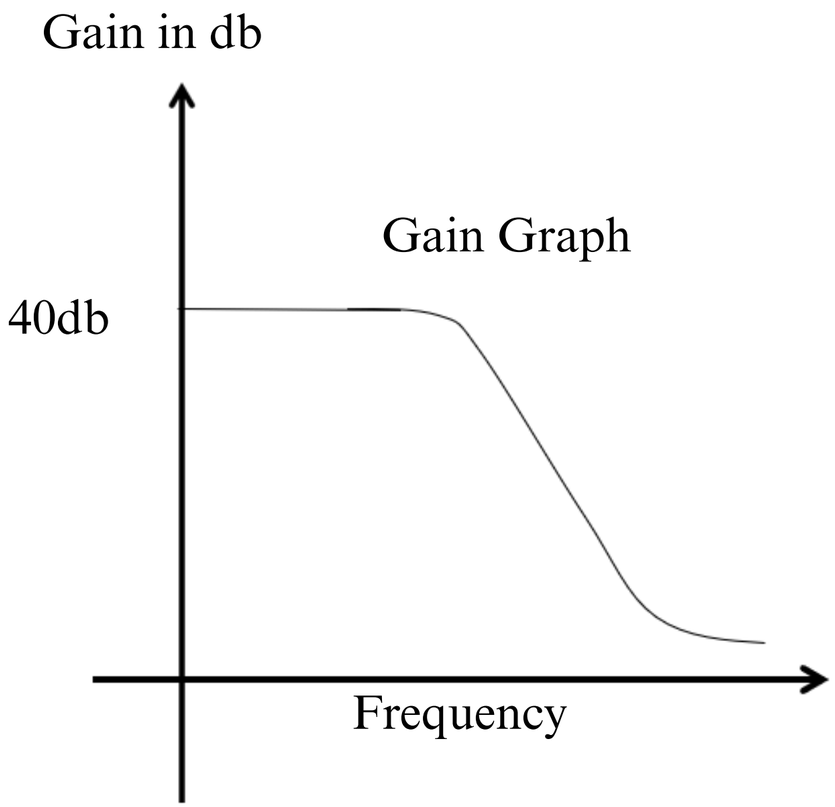
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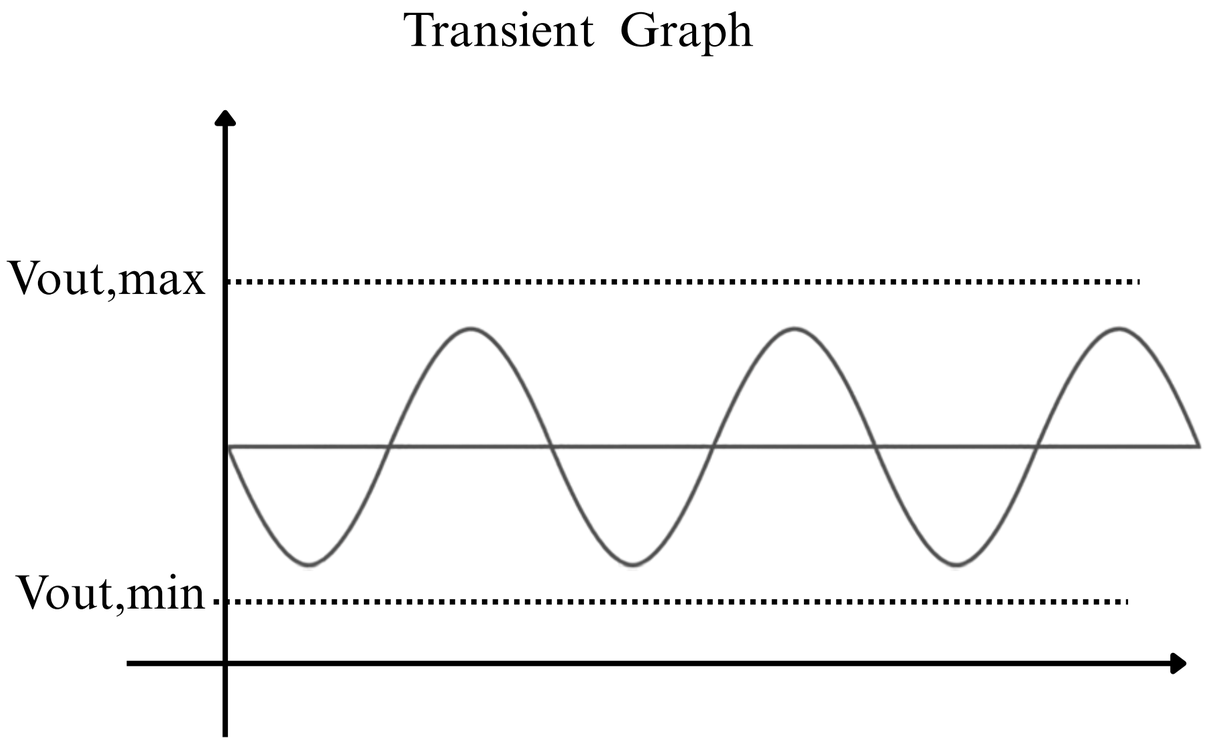
EXPECTED WAVEFORM





EXPECTED WAVEFORM





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Performance

40db	1.2V	80db	36µW
DC Voltage Gain	Output Swing (p-p)	CMRR performance	Power Dissipation

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RESOURCES/REFERENCES USED

- Behzad Razavi , Design Of Analog CMOS Integrated Circuits, Second Edition.
- Analog Integrated Circuit Design , David Johns , Ken Martin, University of Toronto.
- Phillip E. Allen Douglas R. Holberg crvlos Analog Circuit Design Second Edition

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Thank You!

Analog IC Design Mini-Project Presentation